LOKNATH.V- 192421439L

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

Dr.G.Ayyappan

QUESTIONS Total Marks: 10

Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	**1**
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	**1**
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	**1**
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	**1**
5	Functional dependency X → Y means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	**1**
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	**1**
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key	BL2 – Understand	**1**

	reference		
8	If R(A,B,C) has FD: A→B and B→C, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	**1**
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	**1**
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	**1**

Code of Conduct:

*I **LOKNATH.V – 192421439L** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

MCQ					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts

Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ANSWERS

1. Which of the following is true about 1NF?
(a) Allows multi-valued attributes
(b) Eliminates partial dependencies
(c) All attributes must be atomic
(d) Eliminates transitive dependencies

Correct Answer: (c) All attributes must be atomic

Explanation:
1NF (First Normal Form) requires every attribute to contain a **single, indivisible (atomic) value**. It does not allow multi-valued or repeating attributes.

Answer: (c) All attributes must be atomic.

2. A relation is in 2NF if it is in 1NF and:
(a) Has no transitive dependencies
(b) Has no partial dependencies on primary key
(c) Every attribute is a prime attribute
(d) Has no multi-valued dependencies

Correct Answer: (b) Has no partial dependencies on primary key

Explanation:

A relation is in 2NF if it is already in 1NF and has no partial dependency on the primary key.

A partial dependency occurs when a non-key attribute depends only on part of a composite primary key.

For example, if the primary key is (StudentID, CourseID) and StudentID → StudentName, then StudentName depends only on part of the primary key. Therefore, the relation is not in 2NF.

Hence, the correct answer is:

(b) Has no partial dependencies on primary key

3. Which normal form deals with multi-valued dependencies?
(a) 2NF

(b) 3NF
(c) BCNF
(d) 4NF

Correct Answer: (d) 4NF

Explanation:

Fourth Normal Form (4NF) deals with multi-valued dependencies (MVDs).

A relation is in 4NF if, for every non-trivial multi-valued dependency $X \rightarrow\rightarrow Y$, X is a superkey.

Therefore, the correct answer is:

(d) 4NF

5. Functional dependency $X \rightarrow Y$ means:
(a) Y determines X
(b) X determines Y
(c) X and Y are unrelated
(d) Both determine each other

Correct Answer: (b) X determines Y

Explanation:

A functional dependency $X \rightarrow Y$ means that the value of X uniquely determines the value of Y.

Therefore, the correct answer is:

(b) X determines Y

6. Which of the following anomaly is NOT associated with unnormalized relations?

(a) Insertion anomaly

(b) Deletion anomaly

(c) Update anomaly

(d) Retrieval anomaly

Correct Answer: (d) Retrieval anomaly

Explanation:

Insertion, deletion, and update anomalies are common problems associated with poorly designed or unnormalized relations.

Retrieval anomaly is not a standard database anomaly associated with normalization.

Therefore, the correct answer is:

(e) Retrieval anomaly

7.A superkey is:
(a) A minimal set of attributes that uniquely identifies tuples
(b) Any set of attributes that uniquely identifies tuples
(c) An attribute that depends on a non-prime attribute
(d) A foreign key reference

Correct Answer: (b) Any set of attributes that uniquely identifies tuples

Explanation:

A superkey is any set of one or more attributes that can uniquely identify each tuple (row) in a relation.

A candidate key is a minimal superkey, meaning no attribute can be removed without losing uniqueness.

Therefore, the correct answer is:

(b) Any set of attributes that uniquely identifies tuples

8.If R(A,B,C) has FD: A→B and B→C, then C is:
(a) Partially dependent on A
(b) Fully dependent on A
(c) Transitively dependent on A
(d) Independently determined

Correct Answer: (c) Transitively dependent on A

Explanation:

Given the functional dependencies:

$A \rightarrow B$

$B \rightarrow C$

Therefore, A determines C indirectly through B:

$A \to B \to C$

Hence, C is transitively dependent on A.

Therefore, the correct answer is:

(c) Transitively dependent on A

9.5NF is also known as:
(a) Domain-Key Normal Form
(b) (b) Boyce-Codd Normal Form
(c) (c) Project-Join Normal Form
(d) (d) Element Normal Form

Correct Answer: (c) Project-Join Normal Form

Explanation:

5NF is also called Project-Join Normal Form (PJ/NF).

It deals with join dependencies and ensures that a relation cannot be further decomposed into smaller relations without losing information.

Therefore, the correct answer is:

(c) Project-Join Normal Form

10. Lossless-join decomposition ensures:
(a) No data is lost during decomposition
(b) All FDs are preserved
(c) The relation is in BCNF
(d) No null values exist

Correct Answer: (a) No data is lost during decomposition

Explanation:

A lossless-join decomposition ensures that when the decomposed relations are joined back together, the original relation is obtained without losing any information or generating incorrect tuples.

FD preservation and BCNF are separate concepts.

Therefore, the correct answer is:

(a) No data is lost during decomposition